

AMA2111 Mathematics I
3 Ordinary Differential Equation

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A differential equation whose unknown depends on only one independent variable is called an ordinary differential equation (ODE). The unknown of a differential equation will usually be denoted by y and the independent variable by x . For simplicity we shall write y' and y'' , etc. to denote the first and second derivatives of y with respect to x . However, in case of the unknown y being a function of time, we may denote the independent variable by t , and use $\dot{y}$ and $\ddot{y}$, etc. to denote the first and second derivatives of y with respect to t .

Example

If $y(t)$ denotes the position of an object at any time t , then the velocity and the acceleration are respectively given by $v = \dot{y}(t)$ and $a = \ddot{y}(t)$. By Newton's second law, $F = m\ddot{y}$, where F is the force acting on the object and m is the mass of the object. In many examples in mechanics, it is found that F depends on t , y and $\dot{y}$, and the equation of motion for the object is given by the ODE $m\ddot{y} = F(y, \dot{y}, t)$.

A first order differential equation may be written as

$$y' = f(x, y) \quad \text{or} \quad \frac{dy}{dx} = f(x, y).$$

Definition

A real valued function $y = \Phi(x)$, defined on an open interval I , is called a solution of $y' = f(x, y)$ if it satisfies $\Phi'(x) = f(x, \Phi(x))$ for every x in I .

Variables Separable Type

Consider the differential equation

$$\frac{dy}{dx} = g(x)h(y),$$

where $g(x)$ and $h(y)$ are given functions of x and y .

We may now separate the variables x and y and integrate to obtain

$$\int \frac{1}{h(y)} dy = \int g(x) dx.$$

Example 1

Solve the differential equation $e^{2x}y' + e^x = 1$.

Example 2

Solve the differential equation $\frac{dy}{dx} = 2x(1 + y)$.

Example 3

Solve the differential equation $(y + 1)(x^2 + 1) = xy'$.

Reducible to Separable Type

Consider the differential equation

$$\frac{dy}{dx} = G\left(\frac{y}{x}\right),$$

where G is a function with a single variable. Under substitution $v = \frac{y}{x}$, we have

$$y = xv, \quad \frac{dy}{dx} = x \frac{dv}{dx} + v.$$

So v satisfies the ODE

$$x \frac{dv}{dx} + v = G(v), \quad \frac{dv}{dx} = \frac{G(v) - v}{x}.$$

Now we have a variable separable type ODE with unknown function v .

Example 4

Solve the differential equation $\frac{dy}{dx} = \frac{x^2 - y^2}{2xy}$.

1st Order Linear ODE

A 1st order linear ODE has the form

$$\frac{dy}{dx} + p(x)y = q(x),$$

where $p(x)$ and $q(x)$ are given functions of x .

Here we use integrating factor method. Multiply the equation by a function $\mu(x)$, we have $\mu(x)\frac{dy}{dx} + \mu(x)p(x)y = \mu(x)q(x)$. Now if

$$\frac{d\mu}{dx} = \mu(x)p(x),$$

then the equation becomes $\mu(x)\frac{dy}{dx} + \frac{d\mu}{dx}y = \mu(x)q(x)$, or

$$\frac{d(\mu y)}{dx} = \mu(x)q(x).$$

So we have

$$\mu(x)y = \int \mu(x)q(x)dx + C \Rightarrow y = \frac{\int \mu(x)q(x)dx + C}{\mu(x)}.$$

Integrating Factor $\mu(x)$

Note that $\frac{d\mu}{dx} = \mu p(x)$ is a variable separable type 1st order ODE for $\mu(x)$, it is easy to find a integrating factor $\mu(x)$.

$$\int \frac{1}{\mu} d\mu = \int p(x) dx \Rightarrow \ln |\mu| = \int p(x) dx.$$

So one solution is $\mu = e^{\int p(x) dx}$.

The solution of 1st order linear ODE can now be written as

$$y = e^{-\int p(x) dx} \left[\int q(x) e^{\int p(x) dx} dx + C \right].$$

Example 5

Solve the differential equation $\frac{dy}{dx} - \frac{2}{x}y = -\frac{1}{x^2}$.

Bernoulli Equations

A Bernoulli equation is a 1st order ODE in the form

$$y' + p(x)y = q(x)y^n,$$

where $p(x)$ and $q(x)$ are given functions of x and $n \neq 0, 1$.

Note $y = 0$ is a trivial solution. For $y \neq 0$, we divide the equation by y^n to get

$$y^{-n}y' + p(x)y^{1-n} = q(x).$$

Now substitute $v = y^{1-n}$, $v' = (1-n)y^{-n}y'$, the ODE becomes

$$\frac{1}{1-n}v' + p(x)v = q(x),$$

or

$$v' + (1-n)p(x)v = (1-n)q(x).$$

This is a standard 1st order linear ODE for v which we can solve.

Example 6

Solve the differential equation $y' + \frac{4}{x}y = x^3y^2$, $x > 0$.

Initial Value Problem (IVP)

In many applications of differential equations, obtaining the general solution is not enough and we need to find a solution satisfying certain initial conditions. Such a problem is commonly known as an initial value problem (IVP).

To solve the IVP, it is required to find a function $y = \Phi(x)$, defined on a interval I containing the point x_0 , such that

$$\Phi'(x) = f(x, \Phi(x)), x \in I \text{ and } \Phi(x_0) = \alpha.$$

Example 7

Solve the IVP $y' = xy^2$ with $y(1) = 7$.

§2 System of 1st Order Linear ODEs

Consider the system of 1st order ODEs with constant coefficients for n unknown functions of x :

$$\begin{cases} y_1' = a_{11}y_1 + a_{12}y_2 + \cdots + a_{1n}y_n + f_1(x) \\ y_2' = a_{21}y_1 + a_{22}y_2 + \cdots + a_{2n}y_n + f_2(x) \\ \vdots \\ y_n' = a_{n1}y_1 + a_{n2}y_2 + \cdots + a_{nn}y_n + f_n(x) \end{cases}$$

$$\text{Let } A = [a_{ij}]_{n \times n}, \mathbf{y} = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix}, \mathbf{f}(x) = \begin{bmatrix} f_1(x) \\ f_2(x) \\ \vdots \\ f_n(x) \end{bmatrix},$$

The system can be written in matrix form as

$$\mathbf{y}' = A\mathbf{y} + \mathbf{f}(x).$$

Homogeneous ODE System

We only consider the case that the system is homogeneous ($\mathbf{y}' = A\mathbf{y}$) and $A = [a_{ij}]_{n \times n}$ is diagonalizable. Recall that

$$D = \begin{bmatrix} \lambda_1 & 0 & \cdots & 0 \\ 0 & \lambda_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \lambda_n \end{bmatrix}, P = [\mathbf{v}_1, \mathbf{v}_2, \cdots, \mathbf{v}_n]$$

such that $A = PDP^{-1}$.

Use substitution $\mathbf{y} = P\mathbf{w}$, where $\mathbf{w} = \begin{bmatrix} w_1(x) \\ w_2(x) \\ \vdots \\ w_n(x) \end{bmatrix}$ is the column of

new unknown functions. The system $\mathbf{y}' = A\mathbf{y}$ becomes

$$P\mathbf{w}' = AP\mathbf{w} \Rightarrow P^{-1}P\mathbf{w}' = P^{-1}AP\mathbf{w} \Rightarrow \mathbf{w}' = D\mathbf{w}.$$

$\mathbf{w}' = D\mathbf{w}$ can be written as

$$\begin{bmatrix} w'_1 \\ w'_2 \\ \vdots \\ w'_n \end{bmatrix} = \begin{bmatrix} \lambda_1 & 0 & \cdots & 0 \\ 0 & \lambda_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \lambda_n \end{bmatrix} \begin{bmatrix} w_1 \\ w_2 \\ \vdots \\ w_n \end{bmatrix}.$$

We have

$$w'_j = \lambda_j w_j \Rightarrow w_j = c_j e^{\lambda_j x}, \quad j = 1, 2, \dots, n.$$

Therefore

$$\mathbf{y} = P\mathbf{w} = [\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n] \begin{bmatrix} c_1 e^{\lambda_1 x} \\ c_2 e^{\lambda_2 x} \\ \vdots \\ c_n e^{\lambda_n x} \end{bmatrix} = \sum_{j=1}^n c_j \mathbf{v}_j e^{\lambda_j x}$$

is the general solution of the system, where $c_1, \dots, c_n$ are arbitrary constants.

Example 8

Solve the system
$$\begin{cases} y_1' = -y_1 + 2y_2 + 2y_3 \\ y_2' = 2y_1 + 2y_2 + 2y_3 \\ y_3' = -3y_1 - 6y_2 - 6y_3 \end{cases}$$

A second order linear differential equation is an equation of the form

$$y'' + P(x)y' + Q(x)y = R(x),$$

where $P(x)$, $Q(x)$ and $R(x)$ are given functions of x .

In this subject, we shall only treat a special case of the 2nd order ODE where the coefficients of y' and y are independent of x . In other words, we shall consider

$$y'' + py' + qy = R(x),$$

where p and q are real constants. It is called a second order linear differential equation with constant coefficients.

Homogeneous Equation

$$y'' + py' + qy = 0$$

is known as a second order linear homogeneous equation with constant coefficients.

The following facts can be easily established:

- ❶ $y = 0$ is the trivial solution;
- ❷ if $y_1(x)$ and $y_2(x)$ are solutions, then $c_1y_1 + c_2y_2$ is also a solution for any real constants c_1 and c_2 .

Theorem

Let y_1 and y_2 be any two linearly independent solutions of the homogeneous equation. Then every solution y can be expressed as

$$y(x) = c_1 y_1(x) + c_2 y_2(x)$$

for some constants c_1 and c_2 .

Two solutions $y_1(x)$ and $y_2(x)$ are said to be linearly independent if $c_1 y_1 + c_2 y_2 = 0$ implies $c_1 = c_2 = 0$.

Assume $y'' + py' + qy = 0$ has a solution of the form $y = e^{mx}$, where m is a constant to be determined.

Direct substitution gives

$$(m^2 + pm + q) e^{mx} = 0 \Rightarrow m^2 + pm + q = 0$$

Therefore, $y = e^{mx}$ is a solution if and only if m is a root of the quadratic equation. $m^2 + pm + q = 0$ is commonly known as the auxiliary equation of the homogeneous differential equation.

Case 1

Let a and b be the roots of $m^2 + pm + q = 0$.

If a and b be real and distinct, then

$$y_1 = e^{ax} \text{ and } y_2 = e^{bx}$$

are linearly independent solutions.

Therefore, the general solution of the differential equation is given by

$$y = c_1 e^{ax} + c_2 e^{bx}$$

where c_1 and c_2 are arbitrary constants.

Case 2

If $a = b$, then $y_1 = e^{ax}$ is a solution.

Assume $y_2 = v(x)y_1$, where $v(x)$ is a function to be determined. Substituting y_2 into the homogeneous equation, we obtain

$$e^{ax} [(a^2 + pa + q)v + (2a + p)v' + v''] = 0.$$

Because a is a double root of the quadratic equation, $a^2 + pa + q = 0$ and $a = -\frac{p}{2}$. So we have $v'' = 0 \Rightarrow v = Cx + D$, where C and D are arbitrary constants.

In particular, we can choose $v = x$. So $y_2 = xe^{ax}$ is a solution of the homogeneous equation which is independent of y_1 .

Therefore, the general solution of the differential equation is given by

$$y = c_1 e^{ax} + c_2 x e^{ax}$$

where c_1 and c_2 are arbitrary constants.

Case 3

If $a = \alpha + i\beta$ and $b = \alpha - i\beta$, where α and β are real. So the complex solutions are $Y = e^{(\alpha \pm i\beta)x}$. Let

$$Y_1 = e^{\alpha x}(\cos \beta x + i \sin \beta x), \quad Y_2 = e^{\alpha x}(\cos \beta x - i \sin \beta x).$$

Then

$$y_1 = \frac{Y_1 + Y_2}{2} = e^{\alpha x} \cos \beta x, \quad y_2 = \frac{Y_1 - Y_2}{2i} = e^{\alpha x} \sin \beta x$$

are two linearly independent real valued solutions.

Therefore, the general solution of the differential equation is given by

$$y = c_1 e^{\alpha x} \cos \beta x + c_2 e^{\alpha x} \sin \beta x$$

where c_1 and c_2 are arbitrary constants.

Example 9

Solve $y'' - 5y' + 6y = 0$.

Solution:

$$m^2 - 5m + 6 = 0 \Rightarrow m = 2, 3 \Rightarrow y_1 = e^{2x}, \quad y_2 = e^{3x}$$

Therefore, the general solution is $y = c_1 e^{2x} + c_2 e^{3x}$.

Example 10

Solve $y'' - 4y' + 4y = 0$.

Solution:

$$m^2 - 4m + 4 = 0 \Rightarrow m = 2 \Rightarrow y_1 = e^{2x}, \quad y_2 = xe^{2x}$$

Therefore, the general solution is $y = c_1 e^{2x} + c_2 x e^{2x}$.

Example 11

Solve $y'' + 2y' + 2y = 0$.

Solution:

$$m^2 + 2m + 2 = 0 \Rightarrow m = -1 \pm i$$

$$\Rightarrow y_1 = e^{-x} \cos x, \quad y_2 = e^{-x} \sin x$$

Therefore, the general solution is $y = c_1 e^{-x} \cos x + c_2 e^{-x} \sin x$.

Nonhomogeneous Equation

Proposition

$$y'' + py' + qy = R(x),$$

is the nonhomogeneous equation. Let y_p be a particular solution. If y is any solution of the nonhomogeneous equation, then $y - y_p$ is a solution of the homogeneous equation

$$y'' + py' + qy = 0.$$

It is easy to prove by subtracting the following two equations,

$$y'' + py' + qy = R(x),$$

$$y_p'' + py_p' + qy_p = R(x),$$

$$\Rightarrow (y - y_p)'' + p(y - y_p)' + q(y - y_p) = 0.$$

We denote $y_h = y - y_p$ as the solution of the homogeneous equation.

As a consequence of the Proposition, the general solution of

$$y'' + py' + qy = R(x),$$

is given by

$$y = y_h + y_p = c_1y_1 + c_2y_2 + y_p.$$

Pay attention that

$$y_p \neq c_1y_1 + c_2y_2$$

for any c_1, c_2 , as they are the solutions of different equations.

We will introduce two methods to find a particular solution y_p .

Method 1: Undetermined Coefficients

Undetermined coefficients method is good to find a particular solution of a nonhomogeneous equation, when $R(x)$ is the following format.

$R(x)$	y_p guess
$ae^{\alpha x}$	$Ae^{\alpha x}$
$a \cos(\beta x)$	$A \cos(\beta x) + B \sin(\beta x)$
$b \sin(\beta x)$	$A \cos(\beta x) + B \sin(\beta x)$
$a \cos(\beta x) + b \sin(\beta x)$	$A \cos(\beta x) + B \sin(\beta x)$
n th degree polynomial	$A_n x^n + A_{n-1} x^{n-1} \dots + A_1 x + A_0$

In the table, A_i , A and B are unknowns.

If $R(x)$ is a product of functions on the left column in last page, then our guess for y_p is the product of corresponding functions on the right column. For example,

$$R(x) = (a_n x^n + \cdots + a_0) e^{\alpha x} \cos(\beta x)$$

the corresponding y_p is

$$y_p = (A_n x^n + \cdots + A_0) e^{\alpha x} \cos(\beta x) + (B_n x^n + \cdots + B_0) e^{\alpha x} \sin(\beta x)$$

Note that the guess y_p from the table fails when it contains one or more terms being solution of the homogeneous equation. When it happens, we modify the guess by multiplying x and if the problem persists, multiply x again.

After we guess the expression of y_p , we submit y_p to the nonhomogeneous equation to determine the unknown coefficients.

Example 12

Solve $y'' + 4y = x^2 - 5x + 4$.

Example 13

Solve $y'' + 4y' + 4y = 6 \sin 3x$.

Example 14

Solve $y'' + 3y' - 4y = 17e^x$.

Method 2: Variation of Parameters

Assume the homogeneous equation $y'' + py' + qy = 0$ has linearly independent solutions $y_1(x)$ and $y_2(x)$. We look for a particular solution of nonhomogeneous equation $y'' + py' + qy = R(x)$ in the form $y_p(x) = v_1(x)y_1(x) + v_2(x)y_2(x)$, where $v_1(x)$ and $v_2(x)$ are functions of x .

$$y'_p = v_1 y'_1 + v_2 y'_2 + v'_1 y_1 + v'_2 y_2$$

Suppose that $v'_1 y_1 + v'_2 y_2 = 0$, then $y'_p = v_1 y'_1 + v_2 y'_2$.

$$y''_p = v'_1 y'_1 + v'_2 y'_2 + v_1 y''_1 + v_2 y''_2$$

$$\begin{aligned} \text{So } y''_p + p y'_p + q y_p &= v'_1 y'_1 + v'_2 y'_2 + v_1 (y''_1 + p y'_1 + q y_1) + v_2 (y''_2 + p y'_2 + q y_2) \\ &= v'_1 y'_1 + v'_2 y'_2 = R(x). \end{aligned}$$

Now we can see that v'_1 and v'_2 satisfy the linear system

$$\begin{cases} y_1 v'_1 + y_2 v'_2 = 0 \\ y'_1 v'_1 + y'_2 v'_2 = R(x) \end{cases} \Leftrightarrow \begin{bmatrix} y_1 & y_2 \\ y'_1 & y'_2 \end{bmatrix} \begin{bmatrix} v'_1 \\ v'_2 \end{bmatrix} = \begin{bmatrix} 0 \\ R(x) \end{bmatrix}$$

Solving the linear system, we have v_1' and v_2' .

Cramer's rule is one method. We define the Wronskian of y_1 and y_2 by

$$W(y_1, y_2) = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix} = y_1 y_2' - y_1' y_2$$

As $W(y_1, y_2) \neq 0$ (always true for linearly independent y_1 and y_2), we can apply the Cramer's rule to solve the linear system and get

$$v_1' = -\frac{y_2 R(x)}{W(y_1, y_2)}, \quad v_2' = \frac{y_1 R(x)}{W(y_1, y_2)}.$$

Hence we have

$$v_1 = -\int \frac{y_2 R(x)}{W(y_1, y_2)} dx, \quad v_2 = \int \frac{y_1 R(x)}{W(y_1, y_2)} dx.$$

Example 15

Solve $y'' - 2y' + y = \frac{e^x}{x^2 + 1}$.

Definition (Improper integral)

Let $f(t)$ be a real valued function defined for $t \geq a$, where a is a fixed real number. We define the **improper integral** of $f(t)$ by

$$\int_a^{\infty} f(t)dt = \lim_{b \rightarrow \infty} \int_a^b f(t)dt.$$

If the above limit exists and is finite, we say that the improper integral $\int_a^{\infty} f(t)dt$ is *convergent*. Otherwise, the improper integral is said to be *divergent*.

Example 16

For any constant $k > 0$, evaluate $\int_0^{\infty} e^{-kt} dt$.

Example 17

Evaluate $\int_0^{\infty} \cos t dt$ and $\int_0^{\infty} \frac{1}{1+t} dt$.

Integration by Parts

Proposition

If $f(t)$ and $g(t)$ are real valued continuously differentiable functions satisfying $\lim_{t \rightarrow \infty} f(t)g(t) = 0$, then

$$\int_a^\infty f(t)g'(t)dt = -f(a)g(a) - \int_a^\infty f'(t)g(t)dt.$$

Example 18

For any constant $k > 0$, evaluate $\int_0^\infty te^{-kt}dt$.

Definition

For any real valued function $f(t)$ defined for $t \geq 0$, one defines the **Laplace transform** of $f(t)$ by

$$\mathcal{L}\{f(t)\} = F(s) = \int_0^{\infty} e^{-st} f(t) dt.$$

It is an integral transform that converts a function of a real variable t (often time) to a function of a complex variable $s = \sigma + i\omega$ (complex frequency).

Note that $F(s)$ is defined only for those values of s such that the above improper integral converges.

Example 19

Find the Laplace transform of $f(t) = 1$ and $f(t) = t$.

Proposition

It is clear that if $\mathcal{L}\{f(t)\} = F(s)$ and $\mathcal{L}\{g(t)\} = G(s)$, then

$$\mathcal{L}\{af(t) + bg(t)\} = aF(s) + bG(s) = a\mathcal{L}\{f(t)\} + b\mathcal{L}\{g(t)\}$$

for any real constants a and b .

Remark

It is clear that if $f(t) = g(t)$ for $t \geq 0$, then $\mathcal{L}\{f(t)\} = \mathcal{L}\{g(t)\}$. For instance, let $\mathcal{H}(t)$ be the unit step function defined by

$$\mathcal{H}(t) = \begin{cases} 1, & t \geq 0, \\ 0, & t < 0, \end{cases}$$

then $\mathcal{L}\{\mathcal{H}(t)\} = \mathcal{L}\{1\} = \frac{1}{s}$.

Transform of Derivatives

If $f(t)$ and $f'(t)$ are continuous for $t \geq 0$ such that $f(t)e^{-st} \rightarrow 0$ as $t \rightarrow \infty$, we may apply the integration by parts formula to obtain

$$\begin{aligned}\mathcal{L}\{f'(t)\} &= \int_0^{\infty} f'(t)e^{-st} dt = -f(0) - \int_0^{\infty} f(t) \frac{d}{dt} e^{-st} \\ &= -f(0) + s \int_0^{\infty} f(t)e^{-st} dt = sF(s) - f(0).\end{aligned}$$

If we further assume that $f''(t)$ is continuous and $f'(t)e^{-st} \rightarrow 0$ as $t \rightarrow \infty$, then we may obtain

$$\mathcal{L}\{f''(t)\} = s\mathcal{L}\{f'(t)\} - f'(0) = s^2F(s) - sf(0) - f'(0).$$

In general, for any positive integer n , we have

$$\mathcal{L}\{f^{(n)}(t)\} = s^n F(s) - s^{n-1}f(0) - \dots - sf^{(n-2)}(0) - f^{(n-1)}(0)$$

provided the function $f(t)$ has continuous derivatives up to order n , and $f^{(k)}(t)e^{-st} \rightarrow 0$ as $t \rightarrow \infty$ for every $k = 0, 1, \dots, n-1$.

Inverse Laplace Transform

Definition

For a given function $F(s)$, if we can find a function $f(t)$ such that $\mathcal{L}\{f(t)\} = F(s)$, then $f(t)$ is called the **inverse Laplace transform** of $F(s)$, and we denote by

$$f(t) = \mathcal{L}^{-1}\{F(s)\} = \frac{1}{2\pi i} \lim_{T \rightarrow \infty} \int_{\gamma - iT}^{\gamma + iT} e^{st} F(s) ds.$$

Example

Since $\mathcal{L}\{\sin \omega t\} = \frac{\omega}{s^2 + \omega^2}$, we have $\mathcal{L}^{-1}\left\{\frac{1}{s^2 + \omega^2}\right\} = \frac{\sin \omega t}{\omega}$.

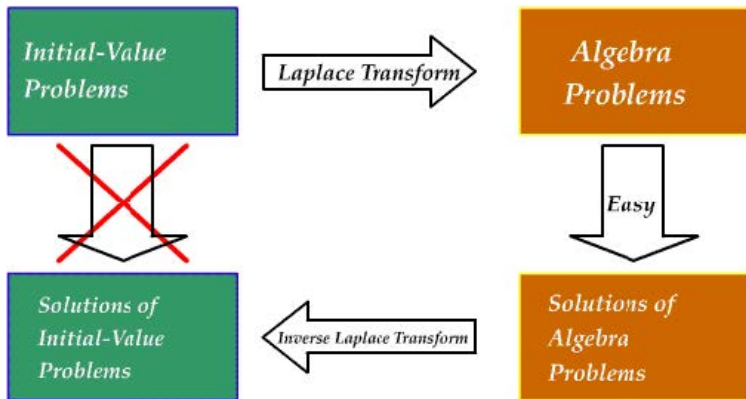
We are not going to use the explicit formula for computing the inverse Laplace transform of a given function of s . Instead, it turns out that with the aid of the *Laplace Transform Table* and some techniques from elementary algebra, we are able to find $\mathcal{L}^{-1}\{F(s)\}$ for a large number of functions.

Example 20

Find $\mathcal{L}^{-1} \left\{ \frac{5s^2 + 3s + 1}{(s^2 + 1)(s + 2)} \right\}$.

Application of Laplace Transform

The most important application of Laplace transform is to solve linear differential equations with constant coefficients.



Consider the initial value problem

$$y''(t) + py'(t) + qy(t) = f(t) \quad (*)$$

with initial conditions $y(0) = \alpha$ and $y'(0) = \beta$.

Putting $Y(s) = \mathcal{L}\{y(t)\}$ and applying Laplace transform on both sides of $(*)$, we obtain

$$s^2 Y(s) - s\alpha - \beta + p[sY(s) - \alpha] + qY(s) = F(s).$$

This is an algebraic equation with unknown $Y(s)$. Its solution is given by

$$Y(s) = \frac{F(s) + \alpha s + \alpha p + \beta}{s^2 + ps + q}.$$

Finally, $y(t)$ may be found by taking inverse Laplace transform.

Example 21

Solve the initial value problem $\frac{dy}{dt} = 4y + 9te^t$ with $y(0) = 1$.

Example 22

For any positive constants ω and k , solve the initial value problem

$$y''(t) + k^2 y(t) = \cos(\omega t)$$

with initial conditions $y(0) = 0$ and $y'(0) = 0$.

This differential equation governs the vibration of a mechanical system without damping under an oscillatory external force.

Transform of a Shift in t

We have defined the unit step function by

$$\mathcal{H}(t) = \begin{cases} 1, & t \geq 0, \\ 0, & t < 0. \end{cases}$$

If $0 \leq a < b$, it is clear that

$$\mathcal{H}(t-a) - \mathcal{H}(t-b) = \begin{cases} 0, & t < a, \\ 1, & a \leq t < b, \\ 0, & t \geq b. \end{cases}$$

Clearly, $\mathcal{L}\{\mathcal{H}(t)\} = \frac{1}{s}$. Moreover, if $a > 0$, then

$$\mathcal{L}\{\mathcal{H}(t-a)\} = \int_0^{\infty} e^{-st} \mathcal{H}(t-a) dt = \int_a^{\infty} e^{-st} dt = \frac{e^{-as}}{s}.$$

Theorem

If $\mathcal{L}\{f(t)\} = F(s)$, then $\mathcal{L}\{f(t-a)\mathcal{H}(t-a)\} = F(s)e^{-as}$.

Example 23

Find the inverse Laplace transform of $\frac{e^{-\pi s}}{s^2 + 4s + 5}$.

Example 24

Solve the initial value problem $y''(t) - y(t) = f(t)$ with $y(0) = y'(0) = 0$ and $f(t) = \begin{cases} t, & 0 \leq t < 2, \\ 0, & t \geq 2. \end{cases}$

Convolution*

Convolution is an operation on two functions f and g , producing a third function that is typically viewed as a modified version of one of the original functions, giving the area overlap between the two functions as a function of the amount that one of the original functions is translated.

Definition

Given two functions $f(t)$ and $g(t)$, we define for any $t \geq 0$,

$$(f * g)(t) = \int_0^t f(x)g(t-x)dx.$$

The function $(f * g)(t)$ is called the **convolution** of $f(t)$ and $g(t)$.

Using the change of variable $x = t - y$ in the definition, we have

$$(f * g)(t) = \int_0^t f(t-y)g(y)d(t-y) = \int_0^t g(y)f(t-y)dy = (g * f)(t).$$

Example

(1) If $f(t) = t$ and $g(t) = 1$, then

$$(f * g)(t) = \int_0^t x \cdot 1 dx = \frac{t^2}{2}.$$

(2) If $f(t) = e^{at}$ and $g(t) = e^{bt}$ where $a \neq b$, then

$$(f * g)(t) = \int_0^t e^{ax} e^{b(t-x)} dx = \frac{e^{at} - e^{bt}}{a - b}.$$

(3) If $f(t) = g(t) = \sin \omega t$, then

$$\begin{aligned}(f * g)(t) &= \int_0^t \sin \omega x \sin \omega(t - x) dx \\ &= \int_0^t \frac{\cos \omega(2x - t) - \cos \omega t}{2} dx = \frac{\sin \omega t - \omega t \cos \omega t}{2\omega}.\end{aligned}$$

Convolution Theorem

Theorem (Convolution Theorem)

If $\mathcal{L}\{f(t)\} = F(s)$ and $\mathcal{L}\{g(t)\} = G(s)$, then

$$\mathcal{L}\{(f * g)(t)\} = F(s)G(s).$$

In other words, we have

$$\mathcal{L}^{-1}\{F(s)G(s)\} = (f * g)(t).$$

Example 25

Find the inverse Laplace transform of $\frac{1}{s^2(s-a)}$ where $a \neq 0$.

Example 26

Solve the differential equation $y''(t) + \omega^2 y(t) = f(t)$ with initial conditions $y(0) = \alpha$ and $y'(0) = \beta$.